BRNO UNIVERSITY OF TECHNOLOGY FACULTY OF ELECTRICAL ENGINEERING AND COMMUNICATION

Robotics and AI

MPC-MAP Assignment No. 2 - Report

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Task 2

For each sensor I used a different map, for a GNSS outdoor map and for a LiDAR indoor map with walls close enough to the sensor.

Channel	1	2	3	4	5	6	7	8
STD (m)	0.0507	0.0493	0.0506	0.0509	0.0488	0.0479	0.0496	0.0516

Tab 1 – Measured standard deviation of LiDAR data from 1000 samples

Axis	x	y
STD (m)	0.4909	0.4895

Tab 2 – Measured standard deviation of GNSS data from 1000 samples

Based on this data we can see that the standard deviation results for LiDAR channels are very similar to each other, although the absolute distances were different. So, it doesn't matter what absolute distance of obstacle, the same applies for GNSS sensor. Histograms are shown in Appendix in figure 1 and 2, and they appear to approximate a normal distribution.

In the project *algorithms/setup.m* contains the selection between measurement source: LiDAR (1) and GNSS (2).

Task 3

Covariance matrix is for a LiDAR 8x8 and for a GNSS 2x2. Covariances between channels/axes are two orders smaller than variances, so they are unrelated between each other. The values on the main diagonal are equal to the variances. The matrices are printed to the console after selecting the measurement source, same as in the previous step.

Task 4

For this task I saved values from the sensors (because of their demanding maps). For a smooth plot the values on the x axis must be sorted. Then the mean value must be subtracted, because we are plotting mean = 0. In Appendix in figure 3 we can see that GNSS is noisier than LiDAR measurement. This gives us more confidence in the LiDAR. To run this function, execute *algorithms/task4.m* as a standalone script.

Task 5

I hardcoded a sequence of motion to achieve the goal position in the map. One of the successful attempts is shown in Appendix figure 4. It is evident that uncertainty exists in the robot's motion, even when both wheels are commanded to rotate at the same speed, the robot does not travel in a perfectly straight line. The potential sources of this uncertainty are: different wheel diameter, wheel slippage, or different motor/shaft parameters. To run the hardcoded sequence, select config number 3 in *algorithms/setup.m*.

Appendix

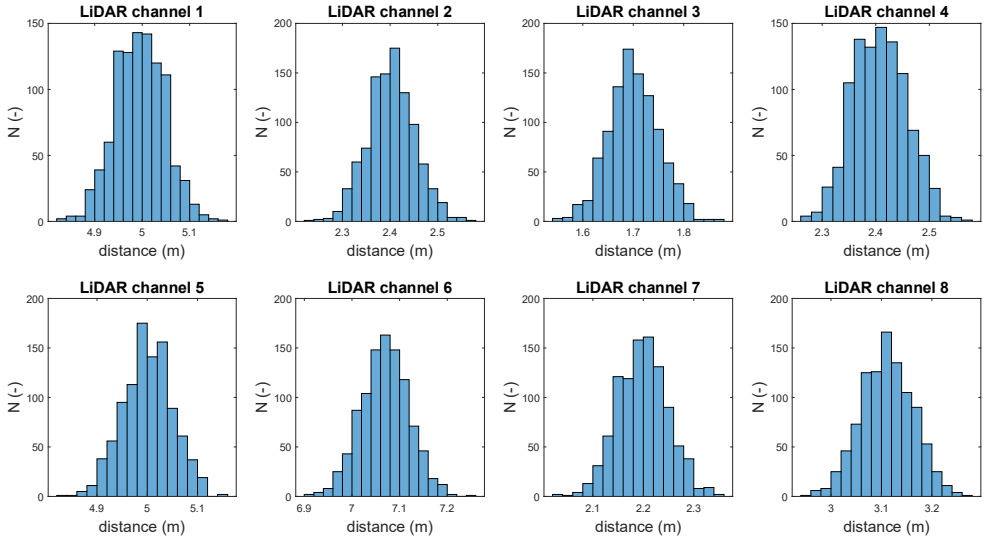

Fig. 1 – Histograms of all the LiDAR channels with measured over 1000 samples

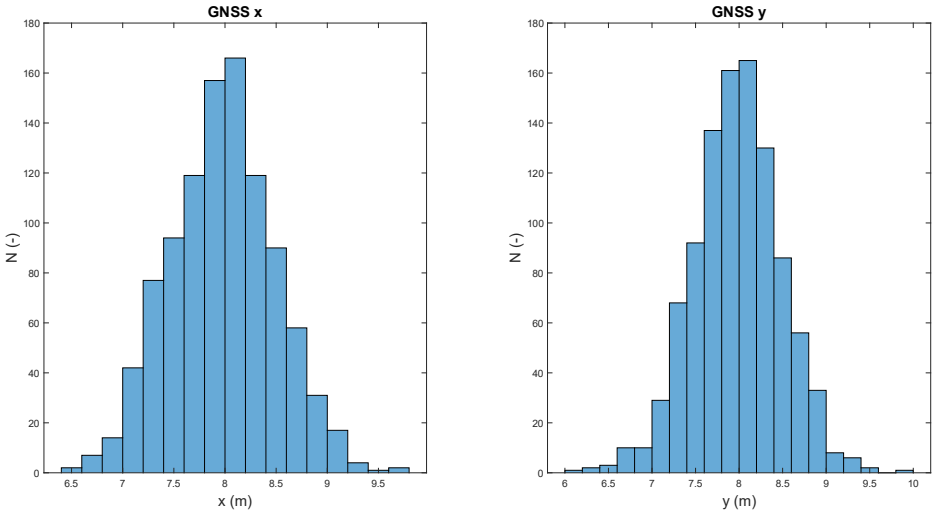

Fig. 2 – Histograms of both GNSS sensor axes measured over 1000 samples

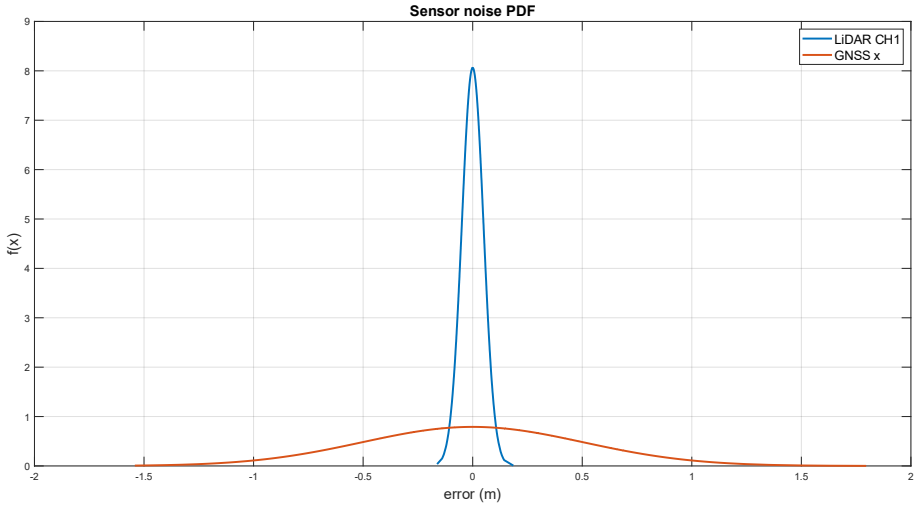

Fig. 3 – Probability density function comparison between two different sensors

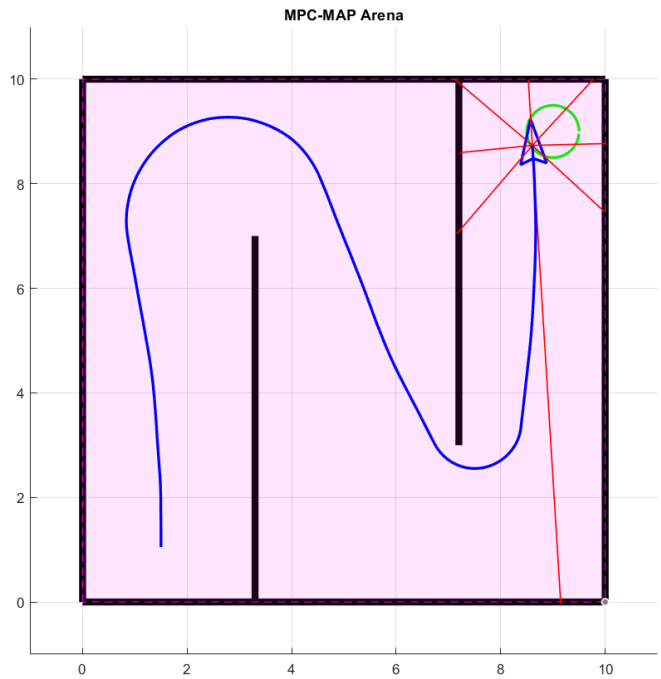

Fig. 4 – Successful run after hardcoded trajectory